Experiment 1

Aim

To design an ER Diagram for an Indian e-commerce platform containing the entities customer, Product, Order, Order Items, Seller, Category, Payment, Delivery and Address and to represent primary keys, composite attributes, multivalued attributes, weak entities, specialization & participation constraints.

① Entities & Attributes

Entity	Attributes
Customer	Customer_ID, Name, Email, Phone.
Address	Address_ID, Address_Details
Product	Product_ID, Product_Name
Seller	Seller_ID, Seller_Name.

② Composite Attributes

For Customer, Use:

Name
- First_Name
- Middle_Name
- Last_Name.

For address, use:

Address_Details.
- House_No
- Street
- Area
- City
- State

③ Multi Volved Attribute

For Customer:

Phone

A customer may have more than one phone number.

Represent it using a double oval:

Customer → Phone

For the relational implement, a seperate table named Customer Phone will be created.

④ Weak Entity

Use Order Item as the weak entity.

→ Identifying Relationship

Order_contains_Order Item

→ Order Item can be identified using:

Order_ID + Product_ID

Therefore:

- Order_ID → Foreign Key
- Product_ID → Foreign Key
- (Order_ID, Product_ID) → Primary Key.

⑤ Speciazation of Product

Specialize Product into:

- Electronics
- Clothing
- Grocery

Common attributes remain in Product.

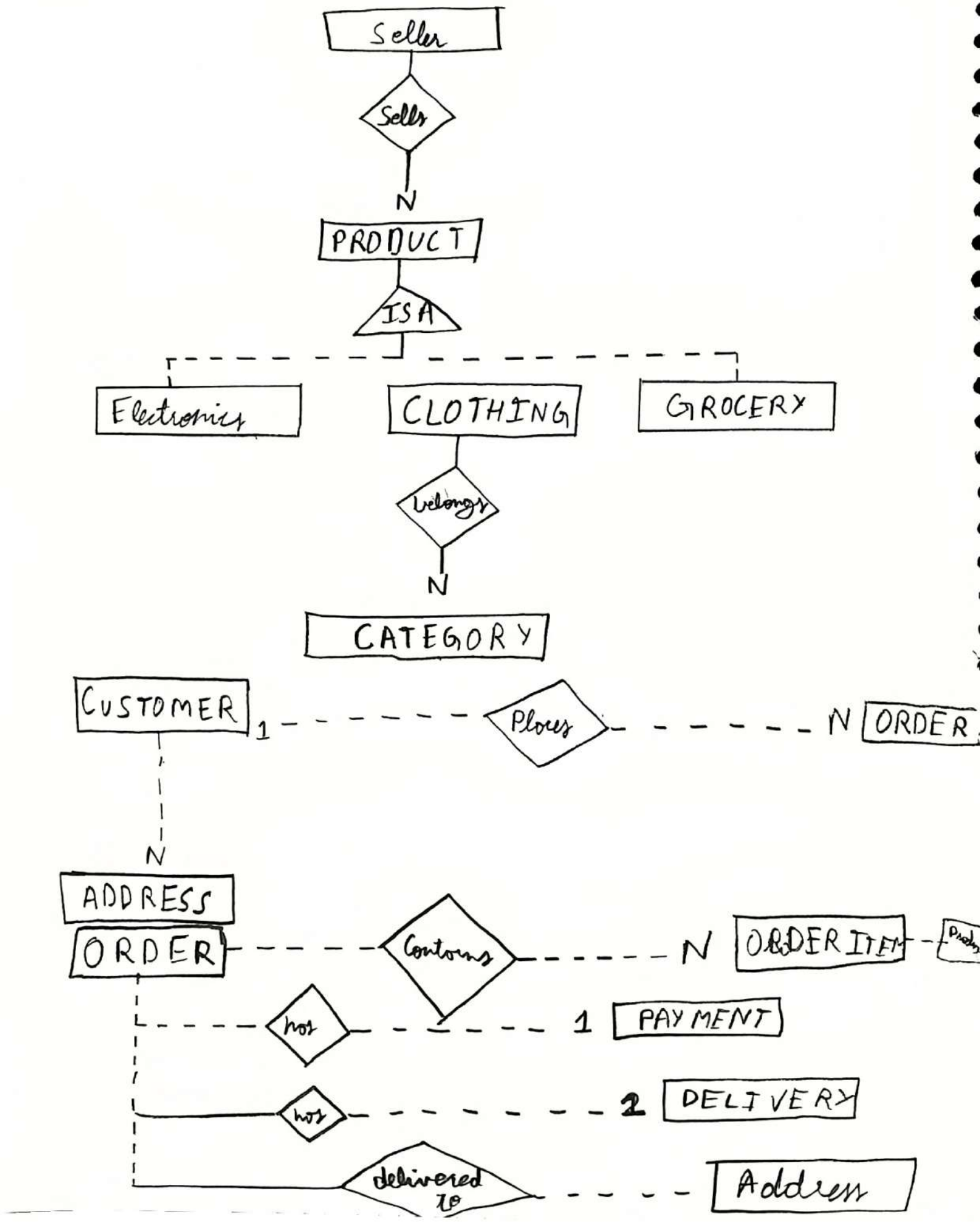
Electronics
- Worranty_Period

Clothing
- Size
- Fabric

Grocery
- Expiry_Date
- Weight

⑥ Relationships

Customer – Places – order	1 : N	Order = Total, Status : Pending
Customer – has – Address.	1 : N	Address : ,, ,,
Seller – sells – Product	1 : N	Product : Total
Category – contains – Product	1 : N	,,
Order – contains – Order Item	1 : N	Order Item : Total
Order – has – Payment	1 : N*	Payment : Total
Order – has – Delivery	1 : 1	Delivery : Total


Seller
Sells
N
PRODUCT
ISA
Electronics
CLOTHING
GROCERY
belongs
N
CATEGORY
CUSTOMER
1
Places
N
ORDER
N
ADDRESS
ORDER
Contains
N
ORDER ITEM
has
1
PAYMENT
has
1
DELIVERY
delivered to
Address